

Recommender System for Gamification and Motivation

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About me

- Ph.D. Student in CS&AI, UniUD & FBK
 - Supervisors: **A. Bucchiarone, K. Roitero**
 - AI-based Models and Tools for Next-Generation Serious Games
- M.S. in Computer Science, UniUD (2024)
 - Design and Development of Software Systems
- B.S. in Computer Science, UniUD (2021)
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About MoDiS



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MoDiS

Motivational Digital Systems



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Background:

- Software Engineering
- AI
- HCI
- Psychology
- Cognitive Science

Topics and Scope of Activity

MoDiS investigates advanced game-based methodologies and techniques supporting **user engagement, motivation and behaviour change**

Gameful Systems Design and User Research

Design frameworks and guidelines
Model-driven motivational system development



Adaptive and Tailored Gameful Systems

Procedural content generation
Dynamic difficulty adjustment
Recommender Systems
Personalized feedback



Validation and Impact Assessment

Monitoring and impact assessment
Player profiling and behavioural analysis
Social influence in motivational systems



Applications and Impact:

- Expertise in the **design, development and operation** of **game-based motivational systems** and **long-running campaigns** targeting different user groups and applied in different domains (Green Transition, Education, Digital Transformation)

Gameful Recommender Systems for Challenge Assignment



Question

What is motivation?



Human Motivation

- Motivation is the “**willingness of doing things**”
- Can be **intrinsic** or **extrinsic**
- Intrinsic motivation represents the **natural inclination** toward *assimilation, mastery, spontaneous interest, and exploration*
- Extrinsic motivation, on the other hand, is driven by considerations of **utility** rather than by the pleasure derived from the activity itself, and is mostly determined by **external factors**, primarily *regulations and incentives*
- **Extrinsic motivation influences intrinsic motivation**



The Self-Determination Theory



Competence

Feel effective in interacting with the environment and to express own capabilities



Autonomy

Experience own's behavior as self-determined and volitional, rather than controlled by externals



Relatedness

Feel a sense of belongingness and connectedness with others, as well as to feel cared for by significant others in one's social environment



Gamification

“the use of typical elements of games, including in-game points, rules, modalities of cooperation or competition, goals, leaderboards, etc., in a non-game context”

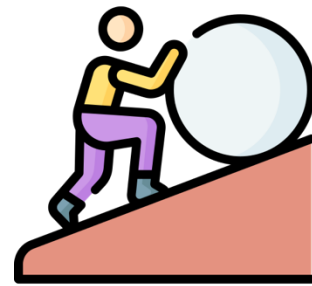


Gamification

- Gamification uses game dynamics to regulate and nudge behaviors and raise extrinsic motivation
- If sustained over time it can raise intrinsic motivation
- One-size-fits-all approaches are not efficacy -> Personalization is needed
- In a Gameful RecSys, personalized in-game actions and elements are assigned to users according to their profile and past interactions.

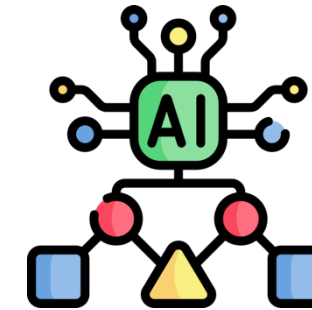
RecSys Applications in Gamification

Dynamic Difficulty Adjustment



Tailoring the game elements difficulty to match user skills

Procedural Content Generation



Automatically generate large amounts of game content, using insights from users' behaviours



RecSys Applications in Gamification

Dynamic Reward Adjustment



Calibrating rewards (virtual or real) based on individual effort and task/challenge difficulty

Fairness and Optimization



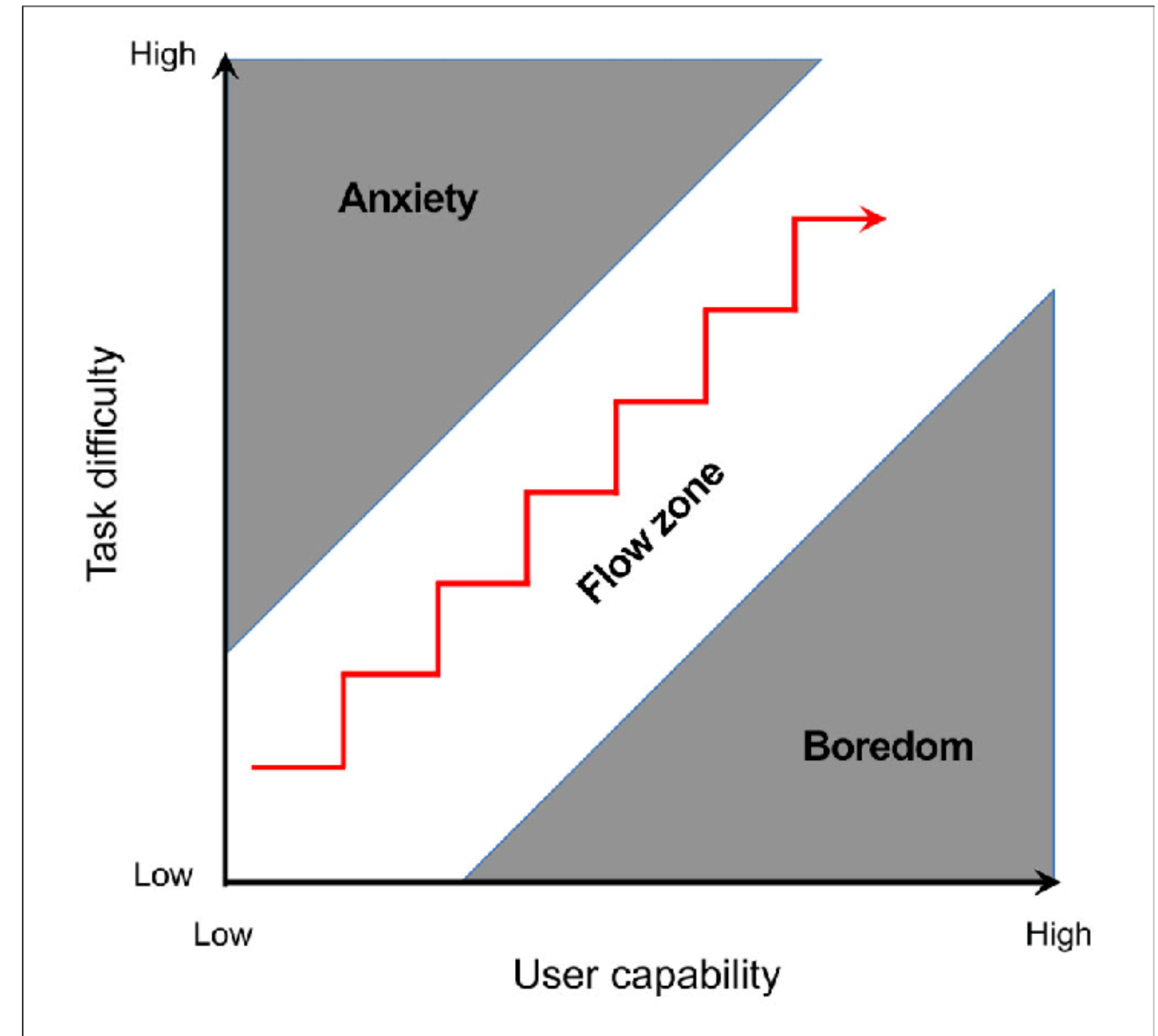
Optimize game dynamics for balanced players experience



The “Flow”

The more the difficulty matches a person's abilities, the more intrinsically motivated they are.

In this context, a RecSys is effective if it keeps the user in a state of “**Flow**” for as long as possible



Feedback Problem in Gameful RecSys

- Usually **completion** feedback (binary) is used
- The score is the **probability of success**
- **Maximize** score -> Easier challenges -> Boredom
- **Minimize** score -> Harder challenges -> Anxiety

Not too easy, not too difficult

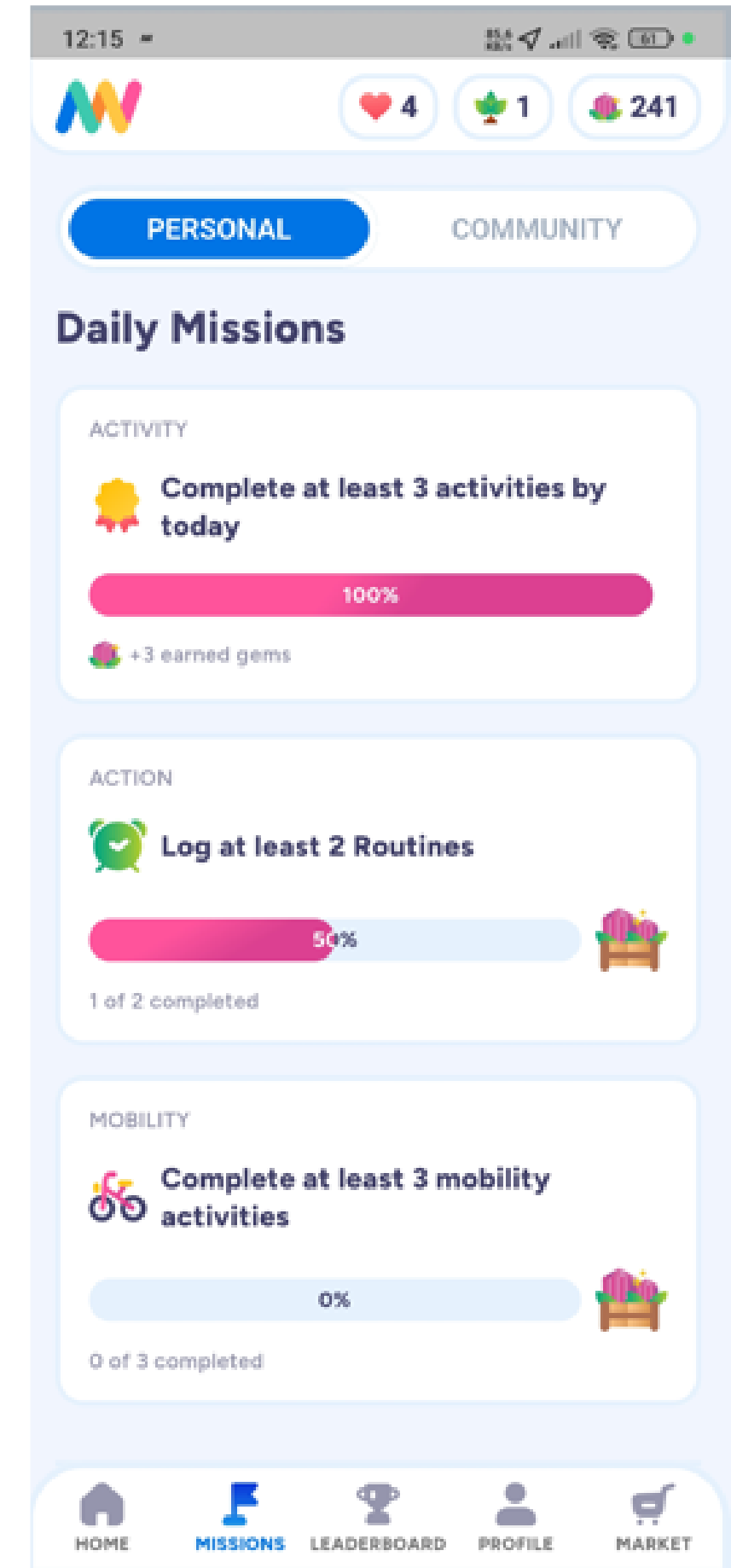


An Example of a Gameful RecSys: AWorld



The Application

- AWorld is a mobile app endorsed by the UN campaign "Act Now" that promotes individual sustainability
- The app uses gamification to educate users, raise awareness, and inspire positive behavioral changes
- It engages users through personalized challenges, known as **missions**, and rewards them with in-game points



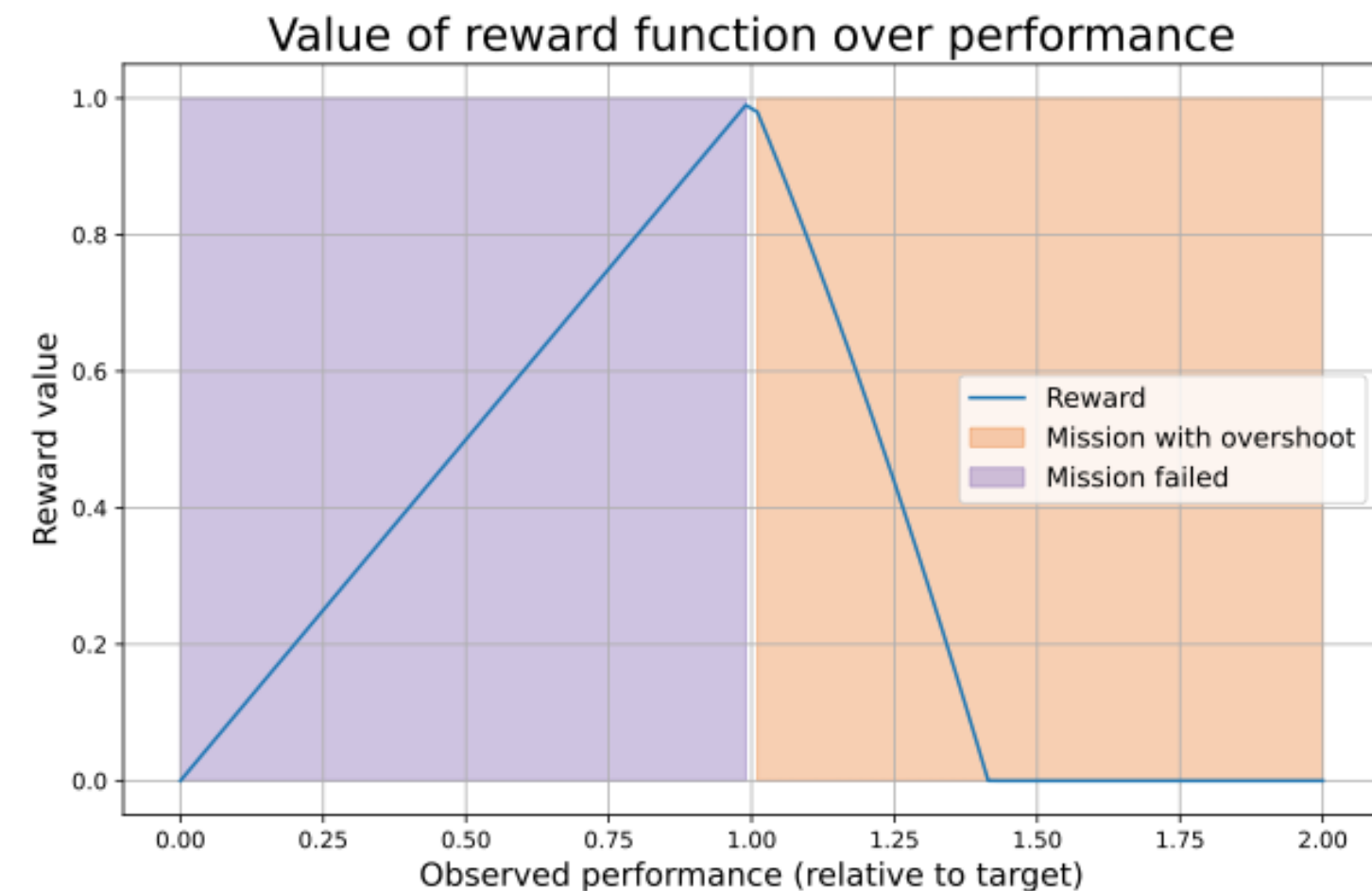
The Business Problem

- Missions in the AWorld App have historically been assigned almost completely at random to users
- The app requires a recommender system to assign missions that match user inclinations while nudging them incrementally to improve their habits
- Missions must be tailored so they are neither too easy nor too hard, matching the user's specific state at that time
- They should also encourage users to explore the app

The Custom Reward Function

- Recognizes partial progress even if a challenge isn't finished
- Reward proportional to the progression -> encouraging gradual improvement
- If a user significantly overshoots a goal, the reward decreases using a quadratic penalty

$$r(p_t(u, m)) = \begin{cases} p_t(u, m) & \text{if } p_t(u, m) \leq 1 \\ \max \{0, 2 - p_t(u, m)^2\} & \text{if } p_t(u, m) > 1 \end{cases}$$



More formally

The cumulated reward R_π received by the policy π after T iterations is

$$R_\pi = \sum_{t=1}^T \sum_{u \in U} \sum_{m \in \pi(u)} r(p_t(u, m))$$

The optimal policy π^* is the one that maximizes the expected value of R_π

$$\pi^* = \arg \max_{\pi} \mathbb{E}[R_\pi] = \arg \max_{\pi} \sum_{t=1}^T \sum_{u \in U} \sum_{m \in \pi(u)} \mathbb{E}[r(p_t(u, m))]$$

The Proposed Solution - MABTree

- MABTree is a hierarchical Multi-Armed Bandit (MAB) framework
- MABTree is designed to replace the app's random mission assignment with a targeted and adaptive approach
- The system utilizes a custom reward function that handles the unique challenges of the app, where tasks involve varying degrees of engagement rather than just binary "completed or not" states

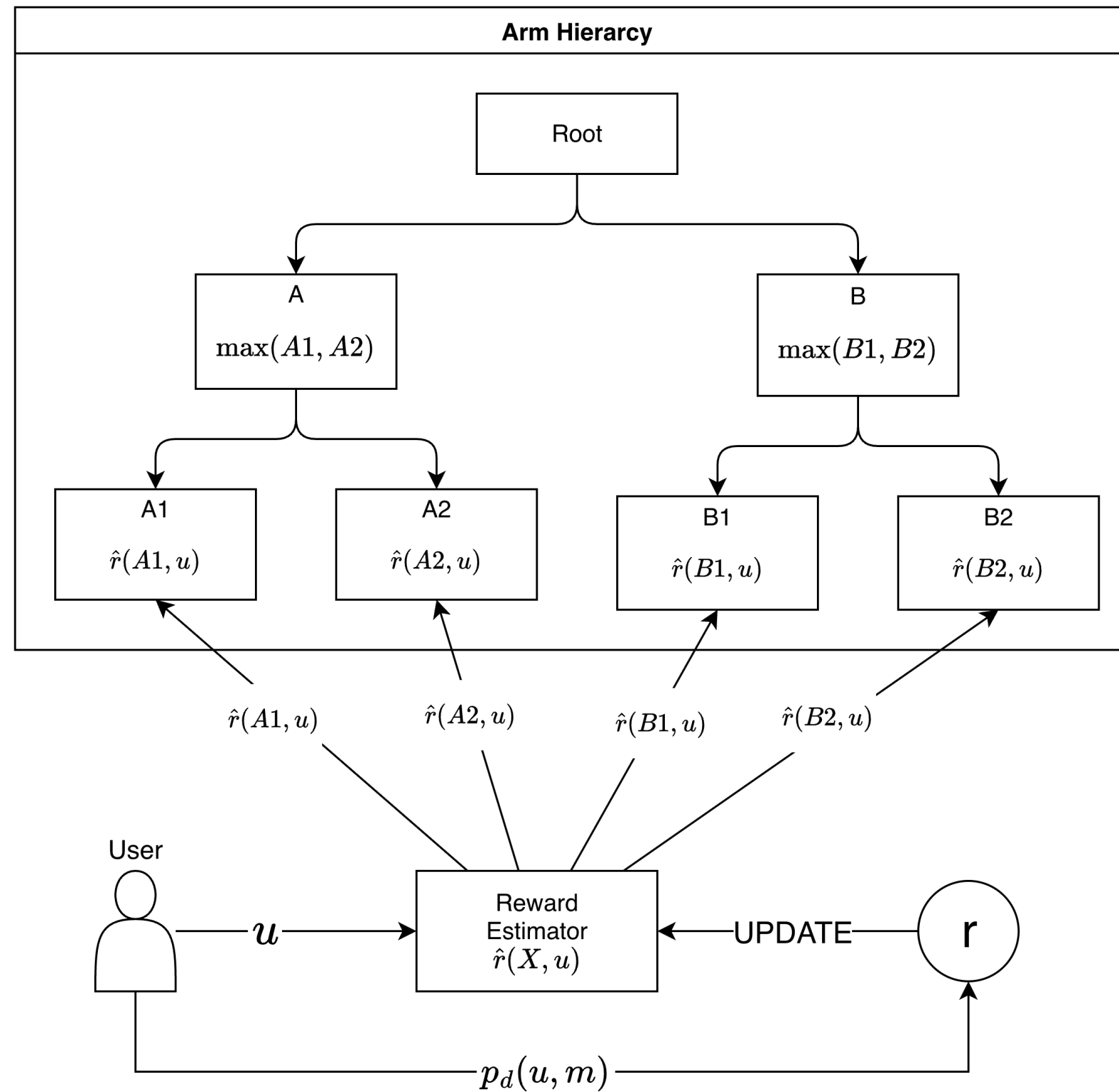
MABTree architecture

- Missions are organized into a tree-like hierarchy grouped by their specific type
- The framework relies on an Arm Hierarchy, a Reward Estimator, and an explore-exploit Policy
- The leaves of the tree represent the actual MAB arms (the missions), while non-leaf nodes represent categories
- The algorithm traverses the tree to make selections using policies like random selection, e-greedy, or UCB1 (Upper Confidence Bound)
- For each leaf and each user, we store the number of times we have assigned it (for UCB1)

Why Tree Bandits?

- Classic MAB assumes that items are **independent**, but
- In an assignment set, *all challenges must be of a different type*
- Given the way the app is designed, *the type of request is more important than the numerical target itself*, but this information depends on the items themselves
- We need to **aggregate** scores of the items to get the score of the types

MABTree architecture



MABTree algorithm

Algorithm 1 MABTree Algorithm for AWorld (Part 1)

```
1: procedure MAIN( $\mathcal{H}, \pi, \mathcal{U}, \hat{r}$ )
2:   for  $t \leftarrow 1, T$  do
3:      $D \leftarrow \emptyset$ 
4:     for all  $u \in \mathcal{U}$  do
5:        $M_u^t \leftarrow \text{SELECT}(\text{ROOT}(\mathcal{H}), \pi, u, \hat{r})$ 
6:       Observe  $p_t(u, m)$  for each  $m_u \in M_u^t$ 
7:        $D \leftarrow D \cup \{\langle u, m, r(p_t(u, m)) \rangle \mid m_u \in M_u^t\}$ 
8:     end for
9:     UPDATE( $\mathcal{H}, \pi, D, \hat{r}$ )
10:  end for
11: end procedure

12: procedure SELECT( $v, \pi, u, \hat{r}$ )
13:  if  $v$  is a leaf node then
14:    return  $\{v\}$ 
15:  else if  $\pi$  is a random policy then
16:     $\mathbb{N} \leftarrow$  Pick  $v.n$  children of  $v$  uniformly at random
17:  else if  $\pi$  is a  $\epsilon$ -greedy policy then
18:     $\mathbb{N} \leftarrow \emptyset$ 
19:     $N \leftarrow \text{CHILDRED}(v)$ 
20:    for  $i \leftarrow 1, v.n$  do
21:      if Random  $< \epsilon$  then
22:         $x \leftarrow$  pick random from  $N$ 
23:      else
24:         $x \leftarrow \arg \max_{v \in N} \text{ESTIMATE}(v, u, \hat{r})$ 
25:      end if
26:       $\mathbb{N} \leftarrow \mathbb{N} \cup \{x\}$ 
27:       $N \leftarrow N \setminus \{x\}$ 
28:    end for
29:  else if  $\pi$  is a UCB policy then
30:     $\mathbb{N} \leftarrow$  Pick  $v.n$  children of  $v$  with the highest  $\text{ESTIMATE}(v, u, \hat{r}) + \sqrt{\frac{\log(t_u)}{\text{COUNT}(v, u)}}$  value
31:  end if
32:  return  $\bigcup_{v \in \mathbb{N}} \text{SELECT}(v, \pi, u, \hat{r})$ 
33: end procedure

34: procedure UPDATE( $\mathcal{H}, \pi, D, \hat{r}$ )
35:  for all  $\langle u, m, r \rangle \in D$  do
36:     $m.c_u = m.c_u + 1$ 
37:     $t_u \leftarrow t_u + 1$ 
38:  end for
39:  Update  $\hat{r}(m, u)$  using  $D$ 
40: end procedure
```

Algorithm 2 MABTree Algorithm for AWorld (Part 2)

```
41: procedure ESTIMATE( $v, u, \hat{r}$ )
42:  if  $v$  is a leaf node then
43:    return  $\hat{r}(v, u)$ 
44:  else
45:    return  $\max_{v' \in \text{CHILDREN}(v)} \text{ESTIMATE}(v', u)$ 
46:  end if
47: end procedure

48: procedure COUNT( $v, u$ )
49:  if  $v$  is a leaf node then
50:    return  $v.c_u$ 
51:  else
52:    return  $\sum_{v' \in \text{CHILDREN}(v)} \text{COUNT}(v', u)$ 
53:  end if
54: end procedure
```

Experimental Setup

- MABTree was evaluated in an offline setting using multiple policies and configurations.
- The evaluation utilized real-world logged mission assignment data from the AWorld App, collected between October and December 2024
- The raw data consisted of approximately 271,000 records
- The dataset covered 13,606 users and 32 distinct missions related to 7 different objectives

Algorithm 3 Replay evaluation

```
1: procedure REPLAY( $\mathcal{H}, \pi, \mathcal{D} = \{\mathcal{D}_1, \dots, \mathcal{D}_T\}$ )  
2:    $h_0 \leftarrow []$  ▷ Empty history  
3:    $M \leftarrow \{M_u \leftarrow \emptyset \mid u \in \mathcal{U}\}$  ▷ Unique missions recommended to each user  
4:   for all  $\mathcal{D}_i \in \mathcal{D}$  do  
5:      $R_i \leftarrow []$  ▷ Missions recommended at current round  
6:     for all unique user  $u \in \mathcal{D}_i$  do  
7:        $M_u^i \leftarrow \text{SELECT}(\mathcal{H}, \pi, u)$   
8:        $M_u \leftarrow M_u \cup M_u^i$   
9:        $R_i \leftarrow \text{CONCAT}(R_i, M_u^i)$   
10:    end for  
11:     $h_i \leftarrow \text{CONCAT}(h_{i-1}, R_i \cap \mathcal{D}_i)$  ▷ Take recommendations which exists in dataset for current round  
12:     $\text{UPDATE}(\mathcal{H}, \pi, h_i)$  ▷ Update history with data collected  
13:  end for  
14:   $R_\pi \leftarrow \sum_{p_t(u,m) \in h_T} r(p_t(u, m))$  ▷ Total reward of all recommendations  
15:   $\text{Cov}_\pi \leftarrow \frac{1}{|\mathcal{U}|} \sum_{M_u \in M} \frac{|M_u|}{|\mathcal{M}|}$  ▷ Average coverage of recommendations  
16:   $S_\pi \leftarrow \frac{1}{T} \sum_{i=1}^T \text{ENTROPY}(R_i)$  ▷ Average entropy of recommendations  
17: end procedure
```

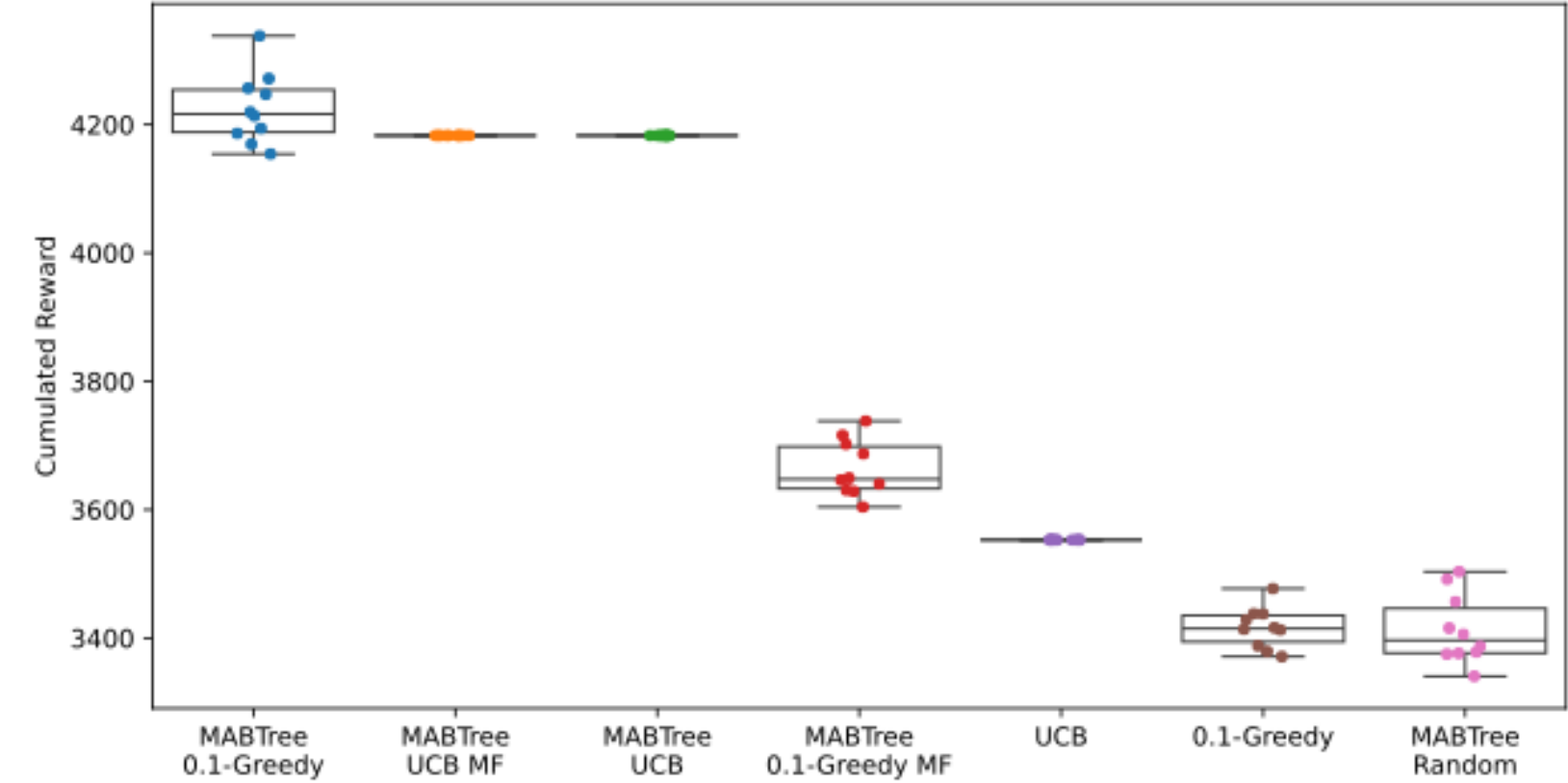
Metrics

- **Cumulated Reward:** sum of the rewards received by the entries in the history that is returned by the replay method
- **Intra-user Coverage:** average proportion of unique missions recommended to each user during the time window considered
- **Intra-round Entropy:** the average Shannon entropy of recommended missions during each round

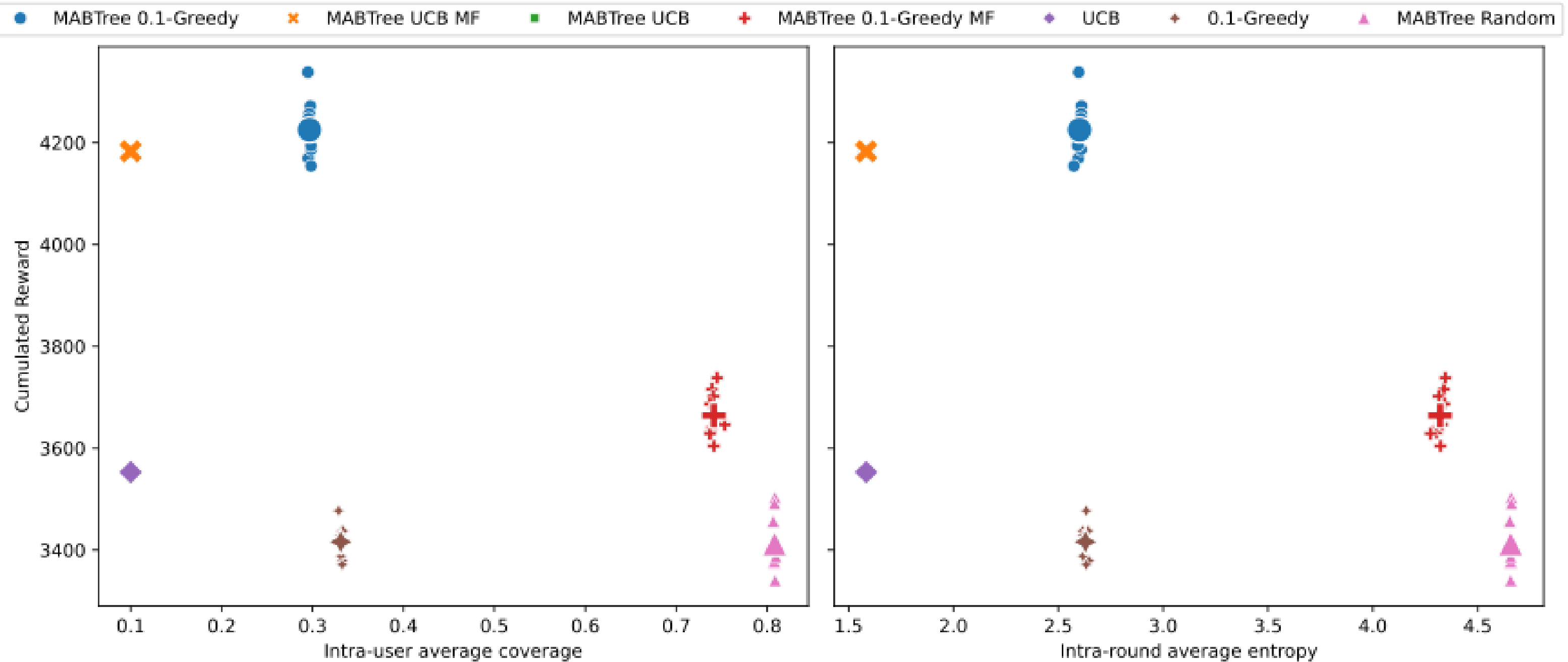
Results

Table 2. Mean and standard deviation of cumulated reward, coverage and entropy for each setup. For each metric, the best and second best setups are highlighted in bold and underlined, respectively. The worst setup is highlighted in italic.

	Entropy		Coverage		Cumulated Reward	
	mean	std	mean	std	mean	std
0.1-Greedy	2.630	0.011	0.331	<u>0.002</u>	3416.241	31.612
MABTree 0.1-Greedy	2.602	<u>0.012</u>	0.297	0.001	4225.157	55.007
MABTree 0.1-Greedy MF	<u>4.320</u>	0.021	<u>0.741</u>	0.005	3664.255	43.747
MABTree Random	4.657	0.002	0.808	0.001	<i>3413.001</i>	<u>54.010</u>
MABTree UCB	1.585	0.000	0.100	0.000	4183.000	0.000
MABTree UCB MF	1.585	0.000	0.100	0.000	<u>4183.000</u>	0.000
UCB	<i>1.585</i>	<i>0.000</i>	<i>0.100</i>	<i>0.000</i>	3553.000	<i>0.000</i>

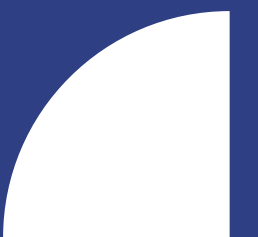


Results



Relationship between Cumulated Reward, Coverage, and Entropy. The plots show the Cumulated Reward versus the Intra-user average coverage (left) and the Intra-round average entropy (right) for the seven algorithms.

Gamification in Collaborative Environments



Collective Gamification

- Collective gamification uses game mechanics and group dynamics to enhance engagement and raise awareness of important issues
- **Aligning personal motivation with collective objectives** is the key to maximizing awareness and driving lasting change

Formalizing the problem

The problem of gamified task assignment is formulated as a mixed-integer linear program (MILP).

$\mathcal{A}$, $\mathcal{T}$ and $\mathcal{S}$ represent the sets of Players, Tasks, and Skills, respectively. A binary Q-matrix $Q \in \{0,1\}^{|\mathcal{T}| \times |\mathcal{S}|}$ maps tasks to their requires skills. Each tasks also carries a global benefit v_t and a reward r_t .

For each player a , the system computes a skill proficiency vector $\hat{a} \in \mathbb{R}^{|\mathcal{S}|}$. An adaptation function $f(\hat{a}, t; \theta_a)$ then evaluates the specific task's value for that player. The objective function uses a weight parameter $\alpha \in [0,1]$ to balance this individual value against the normalized global benefit.

Formalizing the problem

$$\max \sum_{a \in \mathcal{A}} \sum_{t \in \mathcal{T}} V_{at} x_{at}$$

$$V_{at} = \alpha v_t + (1 - \alpha) f(\hat{a}, t; \theta_a)$$

subject to

$$\sum_{t \in \mathcal{T}} x_{at} = n \quad \forall a \in \mathcal{A}$$

$$\sum_{t \in \mathcal{T}} q_{ts} x_{at} \leq K_s^{max} \quad \forall a \in \mathcal{A}, \forall s \in \mathcal{S}$$

$$\sum_{t \in \mathcal{T}} q_{ts} x_{at} \geq K_s^{min} \quad \forall a \in \mathcal{A}, \forall s \in \mathcal{S}$$

$$\sum_{t \in \mathcal{T}} r_t x_{at} \leq R_a \quad \forall a \in \mathcal{A}$$

$$x_{at} \in \{0, 1\} \quad \forall a \in \mathcal{A}, \forall t \in \mathcal{T}$$

The system uses a binary variable $x_{at} = 1$ when task t is assigned to player a . The optimization constraints ensure that:

- Each player is assigned n tasks
- The number of assigned tasks requiring a specific skill remains within defined minimum and maximum bounds
- A player's total accumulated reward is capped at R_a to manage system budgets and ensure equitable distribution

Decentralization

We can divide the global problem into local subproblems, one for each player

$$P_a = \left(\begin{array}{l} \max \sum_{t \in \mathcal{T}} V_{at} x_t \\ \text{subject to} \\ \sum_{t \in \mathcal{T}} x_t = n \\ \sum_{t \in \mathcal{T}} q_{ts} x_t \leq K_s^{\max} \quad \forall s \in \mathcal{S} \\ \sum_{t \in \mathcal{T}} q_{ts} x_t \geq K_s^{\min} \quad \forall s \in \mathcal{S} \\ \sum_{t \in \mathcal{T}} r_t x_t \leq R_a \\ x_t \in \{0, 1\} \quad \forall t \in \mathcal{T} \end{array} \right) \quad \forall a \in \mathcal{A}$$

The optimal solution of each P_a can be combined to obtain optimal solution of the original problem -> Decentralized implementations!

Demonstration

Theorem 1. *Let P^* be an optimal solution to problem P with objective value λ . For each agent $a \in \mathcal{A}$, let P_a^* be an optimal solution to subproblem P_a with objective value λ_a . Then $\lambda = \sum_{a \in \mathcal{A}} \lambda_a$.*

Proof. **Base case:** $\mathcal{A} = \{a\}$ so $P = P_a$ and $\lambda = \lambda_a$. The theorem holds.

Inductive hypothesis: Let $\mathcal{A} = \{a_1, a_2, \dots, a_n\}$ and assume that the theorem holds for $\mathcal{A} \setminus \{a_n\}$.

Inductive step: Let $P_{a_n}^*$ be an optimal solution to subproblem P_{a_n} with objective value λ_{a_n} . Assume for absurd that $\lambda \neq \sum_{a \in \mathcal{A}} \lambda_a$. Then $\lambda = \sum_{a \in \mathcal{A}} \lambda_a + \Delta$ for some $\Delta \neq 0$. For inductive hypothesis $\sum_{a \in \mathcal{A} \setminus \{a_n\}} \lambda_a$ is already maximized. Then the objective value for P_{a_n} is maximized when $\lambda_{a_n} = \lambda_{a_n}^* + \Delta$. But this is a contradiction, because the objective value of P_{a_n} is maximized by hypothesis, so $\Delta = 0$. Therefore $\lambda = \sum_{a \in \mathcal{A}} \lambda_a$. $\square$

An example: mobility challenges

- Three skills: walk, bike, train
- Five challenges: walk 1 Km, walk 5 Km, bike 3 Km, bike 10 Km, train by 30 Km
- Maximum 20 points of reward

	Walk	Bike	Train
W1	1	0	0
W5	1	0	0
B3	0	1	0
B10	0	1	0
T30	0	0	1

	W1	W5	B3	B10	T30
v	0	0.5	0.25	0.75	1
$f(a, t, \theta_a)$	0.5	0.5	1	0.25	0
r	5	10	7	12	15

An example: mobility challenges

- We want to assign 2 challenges to each user
- Max 1 challenge for each type
- At least 1 bike challenge
- $\alpha = 0.4$

An example: mobility challenges

Global scores

Player scores

$$\max 0.4(0.5x_2 + 0.25x_3 + 0.75x_4 + x_5) + 0.6(0.5x_1 + 0.5x_2 + x_3 + 0.25x_4)$$

subject to

$$x_1 + x_2 + x_3 + x_4 + x_5 = 2$$

2 challenges

$$\begin{aligned} x_1 + x_2 &\leq 1 \\ x_3 + x_4 &\leq 1 \\ x_5 &\leq 1 \end{aligned}$$

Max 1 per skill

$$\begin{aligned} x_1 + x_2 &\geq 0 \\ x_3 + x_4 &\geq 1 \\ x_5 &\geq 0 \end{aligned}$$

At least 1 bike

$$5x_1 + 10x_2 + 7x_3 + 12x_4 + 15x_5 \leq 20$$

Max 20 reward

Thank you!

References

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